

WAB

Provadis School of International Management and Technology

Comparing CPython 3.14 JIT, PyPy, and Cython for Smith-Waterman Sequence Alignment

Performance Evaluation and a Robust Cross-Runtime Testing Suite

Rubin Chempananickal James
rubin.chempananickal-james@stud-provadis-hochschule.de
Matriculation Number: D876

Department: Information Technology
Module: Fortgeschrittene Programmierung
Reviewer: Prof. Dr. Henrik Paul

16.03.2026

Abstract

Write your abstract here.

Contents

Abstract	II
Glossary	IV
Abbreviations	V
1. Introduction	1
1.1. Background	1
1.2. Research Questions	1
2. Methods	2
2.1. Design	2
2.2. Data Collection	2
References	i
AI Declaration	ii
Declaration of Authorship	iii

Glossary

Term	Description
Smith-Waterman alignment	A dynamic-programming algorithm for local sequence alignment that computes the highest-scoring matching subsequences between two input sequences.
local alignment	An alignment strategy that searches for the best matching subsections of two sequences instead of forcing an end-to-end alignment.
warm-up phase	The initial execution period in which a JIT-enabled runtime collects profiling information and may compile hot paths before reaching steady-state performance.
benchmark harness	The controlled code and tooling used to execute reproducible performance measurements across implementations and runtimes.
differential testing	A testing strategy that compares the outputs of multiple implementations against the same inputs to detect behavioral deviations.

Abbreviations

Abbreviation	Meaning
JIT	Just-in-Time
DP	Dynamic Programming
SW	Smith-Waterman
CI	Continuous Integration

1. Introduction

Write your introduction here. Alright, maybe I will.

1.1. Background

1.2. Research Questions

2. Methods

Write your methods here.

2.1. Design

2.2. Data Collection

References

- [1] T. F. Smith and M. S. Waterman, “Identification of common molecular subsequences,” *Journal of Molecular Biology*, vol. 147, no. 1, pp. 195–197, 1981, doi: 10.1016/0022-2836(81)90087-5.
- [2] S. Behnel, R. Bradshaw, C. Citro, L. Dalcin, D. S. Seljebotn, and K. Smith, “Cython: The Best of Both Worlds,” *Computing in Science & Engineering*, vol. 13, no. 2, pp. 31–39, 2011, doi: 10.1109/MCSE.2010.118.
- [3] B. Bucher and S. Ostrowski, “PEP 744 – JIT Compilation.” [Online]. Available: <https://peps.python.org/pep-0744/>
- [4] Python Software Foundation, “What's New In Python 3.14.” [Online]. Available: <https://docs.python.org/3/whatsnew/3.14.html>
- [5] The PyPy Team, “PyPy Performance.” [Online]. Available: <https://www.pypy.org/performance.html>

AI Declaration

The usage of AI tools within this project is documented here. I declare that I have documented all interactions with AI tools, including the prompts used and the outputs received.

System	Prompt	Usage
GitHub Copilot 1	I want you to rewrite my template from LaTeX to Typst, and make it look like it did before.	Recreated the template in Typst, maintaining the original appearance.
GitHub Copilot 1	I want you to write some tests Comparing the Python 3.14 JIT Compiler, PyPy, and Cython implementations of the Smith-Waterman algorithm.	Created a test suite that benchmarks the Python 3.14 JIT Compiler, PyPy, and Cython implementations of the Smith-Waterman algorithm.
GitHub Copilot 2	Put the test code in Code/, and use Python 3.15 instead of 3.14.	Added a runtime-generated Smith-Waterman validation and benchmarking suite in Code and retargeted the project text from Python 3.14 to Python 3.15.
GitHub Copilot 3	Okay, use python 3.14. 3.15 is still in pre-release. I've created the conda env named smith and installed the packages	Retargeted the Smith-Waterman suite and Exposé references back to Python 3.14 and validated the project against the configured environment.

Declaration of Authorship

I hereby confirm that I have personally and independently prepared the present work and have not used any sources or aids other than those specified. All passages taken verbatim or in substance from other sources are identified as such. The drawings, illustrations and tables in this work are created by me or have been provided with an appropriate source reference. This work has not been submitted by me to any other university in the same or similar form for the acquisition of an academic degree.

Frankfurt, 16.03.2026

Rubin Chempananickal James